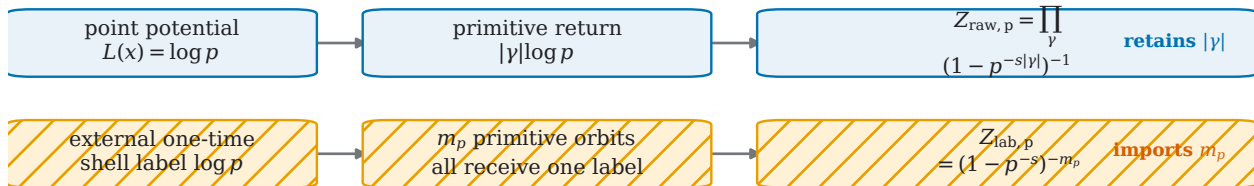


A two different constructions



Dividing a return by its period changes the label; it does not remove any denominator factor.

B ramified stress test at $p = 5$

$$Z_{\text{raw}, 5} = (1 - 5^{-2s})^{-2} (1 - 5^{-10s})^{-2}$$

2 cycles of length 2; 2 cycles of length 10

$$Z_{\text{lab}, 5} = (1 - 5^{-s})^{-4}$$

one label on each of $m_5 = 4$ primitive cycles

Raw mixed factor $\neq$ relabeled degree-four factor.
The constructions answer different questions.

C orbit-label coefficient m_p/r

$p = 2$	1	1/2	1/3
$p = 3$	2	1	2/3
$p = 5$	4	2	4/3
$p = 7$	6	3	2
$p = 11$	24	12	8
	$r = 1$	$r = 2$	$r = 3$

formal repetition (annotations are exact)

All entries are symbolic exact controls; no numerical value of s or $\log p$ is evaluated.